

WT : EXPERIMENT 5

Aim : Simulation of Wireless notice board using tinkercad

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Theory :

A smart attendance system using the NodeMCU ESP8266 microcontroller and an RFID module offers an efficient and automated method for tracking attendance in educational institutions and organizations. This system leverages RFID technology to identify individuals and records their attendance data, which can then be stored and managed digitally.

Working Principle:

A smart attendance system using the NodeMCU ESP8266 microcontroller and an RFID module operates by integrating RFID technology with Wi-Fi connectivity to automate attendance tracking.

1. **RFID Tag Detection:** When an individual places their RFID card or tag near the RFID reader (e.g., MFRC522 module), the reader detects and reads the card's unique identification number.
2. **Data Transmission to NodeMCU:** The RFID reader transmits the unique ID to the NodeMCU ESP8266 microcontroller via the SPI (Serial Peripheral Interface) communication protocol.
3. **Processing and Identification:** The NodeMCU processes the received unique ID to identify the corresponding individual by comparing it against a predefined database of authorized IDs stored within the system.
4. **Attendance Logging:**
 - i) **Timestamp Generation:** Upon successful identification, the NodeMCU generates a timestamp marking the exact date and time of the attendance event.
 - ii) **Data Compilation:** The system compiles the individual's identification details along with the timestamp to create a comprehensive attendance record.
5. **Data Transmission to Online Platform:**

i) Wi-Fi Connectivity: Leveraging the ESP8266's Wi-Fi capabilities, the NodeMCU establishes a connection to a specified online platform, such as Google Sheets or a dedicated web server.

ii) Record Uploading: The compiled attendance data is transmitted and recorded in real-time on the online platform, facilitating immediate access and remote monitoring.

6. User Feedback (Optional): An LCD display or LED indicator can be integrated to provide immediate visual feedback to the user, confirming the successful recording of their attendance or indicating errors if the RFID tag is unrecognized.

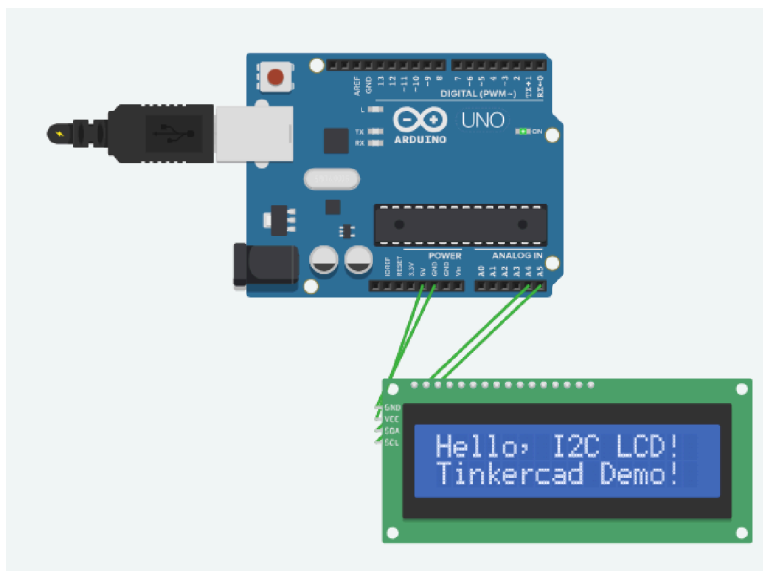
Code :

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
// Set the I2C address (usually 0x27 or 0x3F)
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
  lcd.begin(16, 2);
  lcd.init(); // Initialize the LCD
  lcd.backlight(); // Turn on the backlight
  lcd.print("Hello, I2C LCD!");
}

void loop() {
  lcd.setCursor(0, 1);
  lcd.print("Tinkercad Demo!");
  delay(1000);
}
```

Output :



Conclusion:

Implementing a smart attendance system using the NodeMCU ESP8266 microcontroller and an RFID module offers an efficient and automated solution for tracking attendance in various settings, such as educational institutions and workplaces. This system streamlines the attendance process, reduces manual errors, and provides real-time data access. By leveraging the Wi-Fi capabilities of the ESP8266, attendance records can be seamlessly transmitted to online platforms, facilitating remote monitoring and integration with other management systems. Overall, this approach enhances operational efficiency and ensures accurate record-keeping.